

Republic of the Philippines

Laguna State Polytechnic University

Province of Laguna

ITEP 414 - System Administration and Maintenance

Professional Ubuntu Server Installation Manual

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Portfolio Project: Junior System Administrator Deployment Documentation

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1. Introduction

This manual is intended for an administrator who needs to reproduce the Week 3 Ubuntu Server deployment for ABC Startup Solutions. It converts the course requirements into a practical installation, verification, troubleshooting, and handover procedure. The server is intended as a baseline platform for future file sharing, remote administration, database hosting, web hosting, and internal services.

The manual deliberately does not invent machine-specific evidence. The actual DHCP address, partition scheme, filesystem, command output, and screenshots must be recorded from Cedrick Gayoso's real virtual machine.

2. Deployment Standard

Item	Standard
VM name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB VDI/VMDK
Network	NAT, or Bridged if instructed
Optical drive	Ubuntu Server LTS ISO - latest stable version
Language	English
Keyboard	Appropriate layout, e.g. English (US)
Network configuration	DHCP unless instructed otherwise
Hostname	server01
User	Cedrick Gayoso; documented username gayoso - verify actual VM
Disk method	Guided - Use Entire Disk
SSH	Install OpenSSH Server
Additional packages	None unless instructed

3. Ubuntu Server Installation Guide

3.1 Create the virtual machine

Create a new VM named **Ubuntu-Server-Week03**. Assign 4 GB RAM, 2 virtual processors, a 40 GB VDI/VMDK disk, NAT networking (or Bridged if instructed), and attach the Ubuntu Server ISO.

3.2 Language

Start the installer and select **English**.

3.3 Keyboard Layout

Select the appropriate keyboard layout, such as English (US).

3.4 Network Configuration

Accept DHCP unless instructed otherwise. Record the assigned IP address.

3.5 Hostname and User Account

Set the hostname to **server01**. Create a non-root administrative user. This portfolio uses **gayoso** based on the module username example; the actual VM value should be verified. Use a strong password.

3.6 Disk Partition

Choose **Guided - Use Entire Disk**. Record the actual partition scheme, filesystem used, and 40 GB disk size shown by the installer.

3.7 SSH Server

Enable **Install OpenSSH Server**.

3.8 Additional Packages

Do not install additional packages unless instructed.

3.9 Complete Installation

Finish the installation, restart the virtual machine, and remove the installation ISO if prompted.

4. Initial Server Configuration

4.1 Confirm identity and network

Log in with the non-root administrative account. Confirm the hostname with `hostname` and record the assigned IP address with `ip addr`.

4.2 Apply system updates

Run `sudo apt update` followed by `sudo apt upgrade -y`.

4.3 Confirm SSH service

Run `systemctl status ssh`. The Week 3 module expects the service to show **active (running)**.

4.4 Remote administration

After recording the actual IP address, an administrator can use an SSH client with the documented username and address. Do not publish passwords or private keys.

5. Server Verification Procedures

Check	Command / action	Required result
Login	Local login	Successful login using created account
Hostname	<code>hostname</code>	server01
IP address	<code>ip addr</code>	Record actual DHCP-assigned address
Internet	<code>ping -c 4 google.com</code>	Successful response
Updates	<code>sudo apt update</code> ; <code>sudo apt upgrade -y</code>	Update process completes
SSH	<code>systemctl status ssh</code>	active (running)

6. Troubleshooting Guide

Symptom	Likely area	Action
VM starts installer again	Boot media	Detach/remove the ISO after installation and boot from the virtual disk.
No IP address	Virtual network / DHCP	Check NAT/Bridged mode, virtual NIC connection, and DHCP availability.
Ping fails by name	DNS or connectivity	Check interface status with <code>ip addr</code> ; test gateway/network; verify DNS resolution.
APT update fails	Network, DNS, repository access	Verify connectivity and retry the update commands; record the actual error before changing configuration.
SSH service not found or inactive	OpenSSH installation/service	Confirm OpenSSH Server was installed; inspect <code>systemctl status ssh</code> ; install/start only as allowed by the activity.
Cannot SSH from host	VM network path	Confirm the correct VM IP and network mode; NAT may require port forwarding depending on the hypervisor setup.

7. Best Practices

- Use a non-root administrative account and a strong unique password.
- Install only the packages required for the server role.
- Keep the system updated and document maintenance actions.
- Use SSH for remote administration and protect authentication credentials.
- Record the actual hostname, IP address, disk layout, filesystem, and service status.

- Keep screenshots focused on the required evidence and remove sensitive information.
- Maintain documentation so another administrator can reproduce the deployment.
- Use troubleshooting evidence - commands, logs, and observed errors - before changing configuration.

8. Handover Checklist

- Ubuntu Server installed
- Hostname server01 configured
- Keyboard and network configured
- 40 GB disk configured with Guided - Use Entire Disk
- Actual partition scheme and filesystem recorded
- OpenSSH installed and running
- Login verified
- Actual IP address recorded
- Internet connectivity verified
- System updated
- Required screenshots stored in screenshots/
- Windows Server Evaluation bring-home activity completed
- Installation Guide and Professional Manual exported to PDF
- README updated
- GitHub repository updated
- LinkedIn reflection published

9. References

- Laguna State Polytechnic University, ITEP 414 - System Administration and Maintenance, Week 3 Portfolio Project: Enterprise Server Deployment and Operating System Installation.
- Official Ubuntu Server documentation.
- Official hypervisor documentation for VirtualBox or VMware Workstation.
- Official Microsoft Windows Server documentation for the bring-home activity.